

DEPLOYMENT ACTIVITY 1 REPORT

Student Details:

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Unit Details:

- **Unit Code & Name:** SWE40006 Software Deployment and Evolution
- **Semester:** 9
- **Submission Date:** 28/01/2026

Task Level Attempted:

- Distinction (Task 1.3)

Part 1: Application Development (Task 1.1 & 1.2)

Description:

To fulfill the requirements for Credit (Task 1.2), I developed a custom C# Console Application named **MyApp**. To prepare for the Distinction task, I utilized a modular architecture by creating a separate Class Library (DLL) project named **MyLibrary**. The main application references this library and calls a specific method to display a message, demonstrating successful dependency management between an executable and a dynamic link library.

Evidence 1: Application Source Code

Below is a screenshot of the Visual Studio solution showing the code structure, where **Program.cs** calls the function from **MyLibrary**.

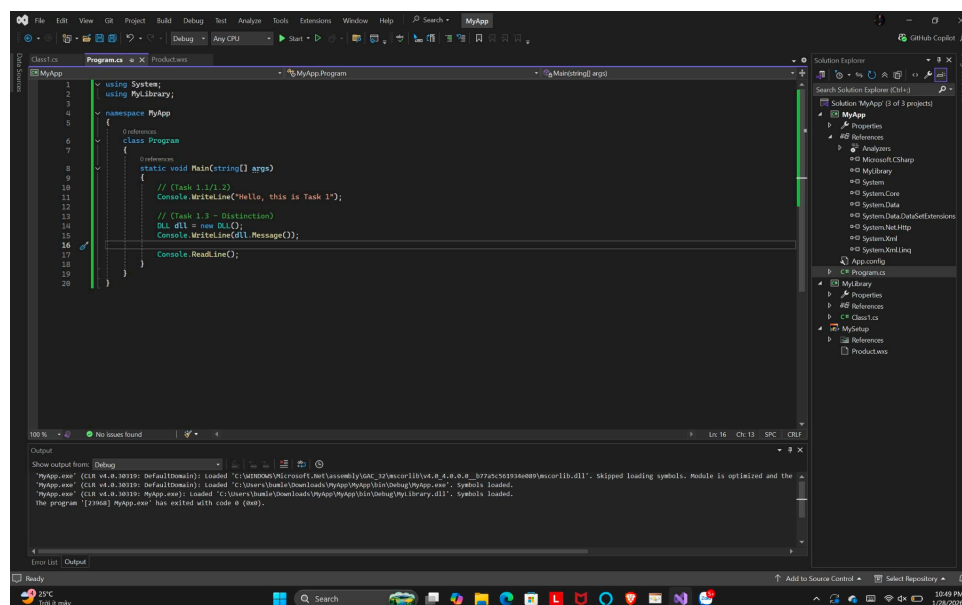


Figure 1: Source code showing the dependency between the main app and the library.

Evidence 2: Application Execution (Development Environment)

Below is the application running successfully within Visual Studio before deployment.

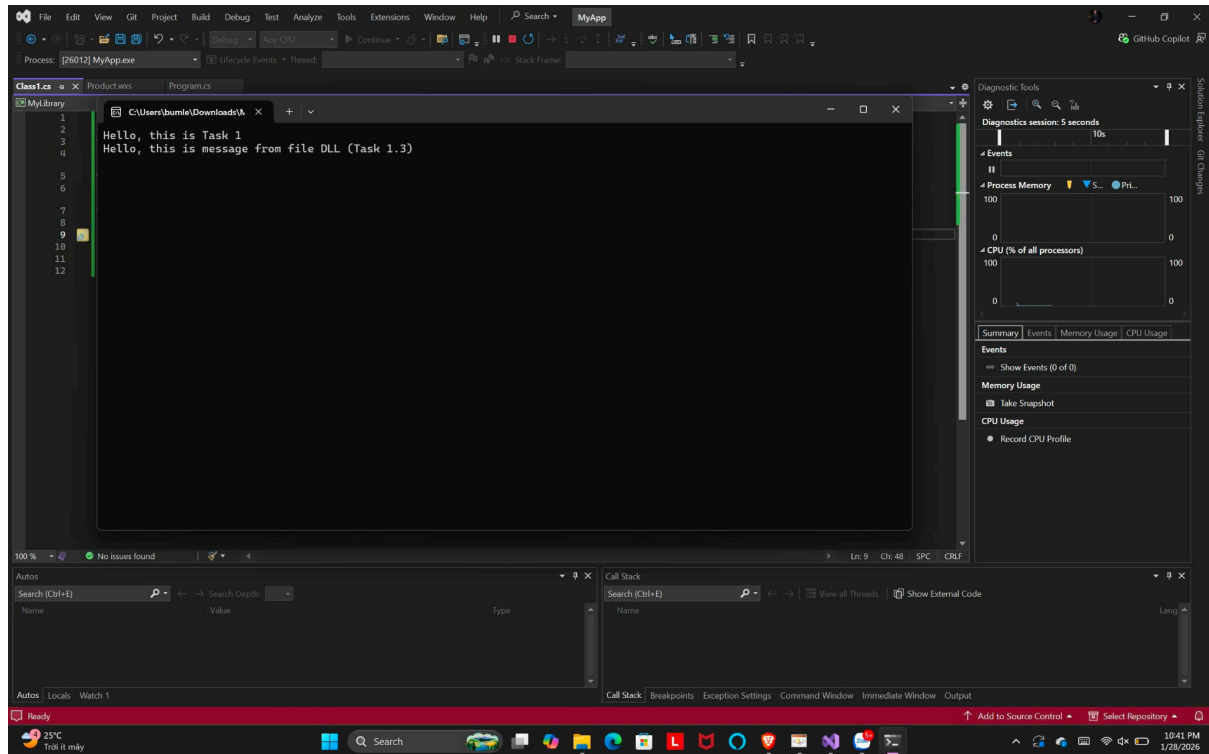


Figure 2: The application running in the development environment displaying the output.

Part 2: WiX Deployment Configuration (Task 1.3 - Distinction)

Description:

For the Distinction task, I utilized the WiX Toolset (v3) to create a Windows Installer (.msi). The `Product.wxs` file was configured to define the directory structure and manage the components. Specifically, I defined two separate `Component` blocks: one for `MyApp.exe` and another for `MyLibrary.dll`. This ensures that the installer correctly packages and deploys the dependency alongside the main executable.

Evidence 3: WiX XML Configuration

Below is the configuration in `Product.wxs` showing the `ComponentGroup` with unique GUIDs for both the executable and the DLL.

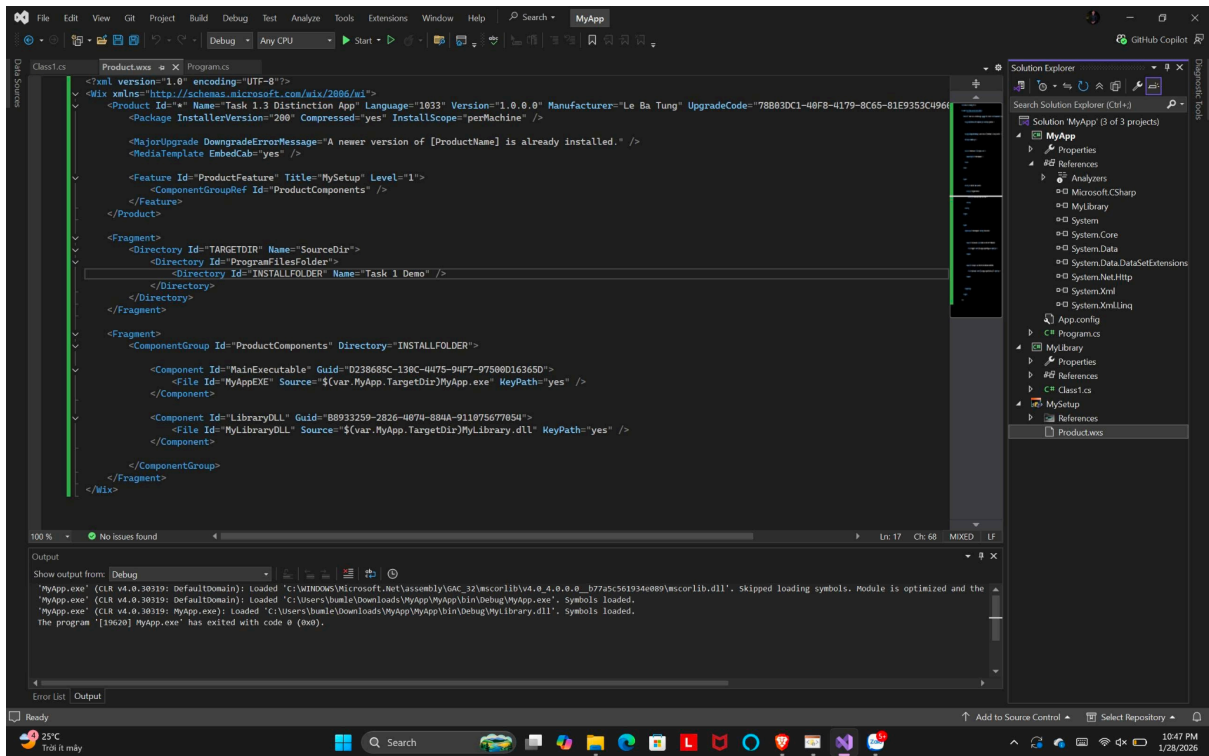


Figure 3: WiX XML configuration defining components for the Exe and Dll files.

Evidence 4: Installation Process

Below is the installer interface running on the target machine.

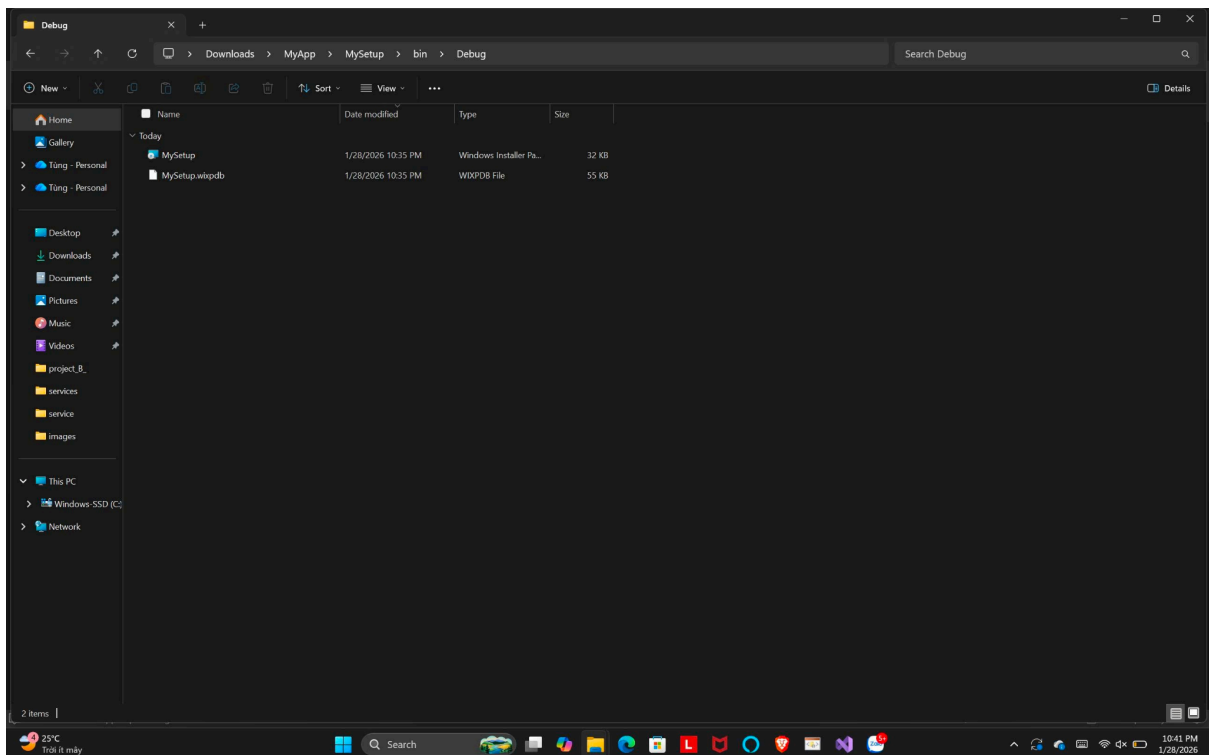


Figure 4: The WiX-generated MSI installer in progress.

Evidence 5: Post-Installation Verification

Below is the installation directory (C:\Program Files (x86)\Task 1 Demo) confirming that both files were deployed successfully.

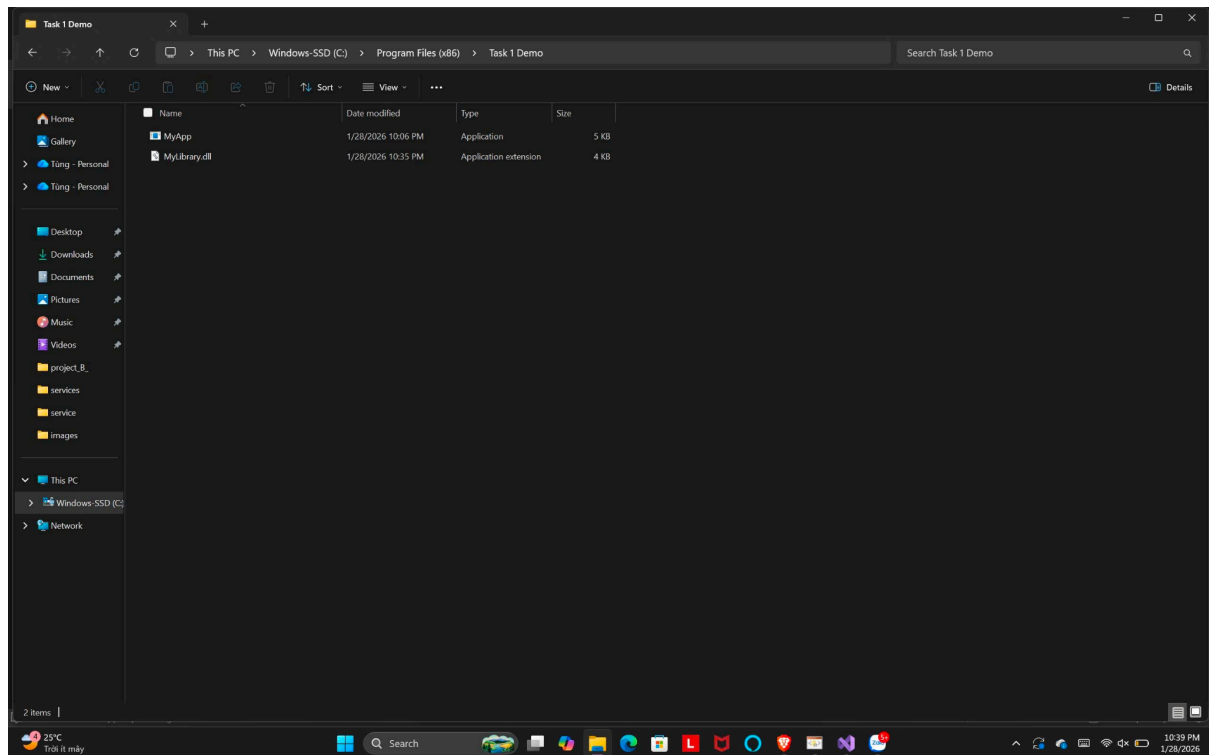


Figure 5: Verification of the installation folder containing both MyApp.exe and MyLibrary.dll.

Source Code Verification

The full source code and WiX project files are available at the link below for verification:

URL: <https://drive.google.com/file/d/1eez19kdReooL8nPvvpAMJJ51qRGAF0em/view?usp=sharing>